



Multi-Scale Spectrogram-Transformer WGAN-GP for Synthetic Signal Generation

Awantika Krishna · Binwant Kaur · Praveena P

AI/ML for Cybersecurity | Faculty of Engineering and Technology | South Asian University, New Delhi

1 – MOTIVATION

Classifiers for medical and audio signal analysis need large labelled datasets. Abnormal signals — arrhythmia beats, unusual sounds — are rare and hard to collect. MIT-BIH ECG has a **5:1 imbalance** between normal and abnormal beats, causing classifiers to miss critical cases.

WHY EXISTING TIME-DOMAIN GANS FAIL

König et al. (2024) use an LSTM to generate synthetic signals directly in time. This produces signals with wrong energy, incorrect statistics, and no frequency structure — the generator cannot see the frequency content it should be matching.

Our approach: generate signals in the frequency domain using a Vision Transformer. Reconstruct the signal in one step using phase learned from real data — no iterative guessing needed.

2 – BACKGROUND

WGAN-GP (GULRAJANI ET AL., 2017)

Uses Wasserstein distance as the training loss — more stable than standard GANs, avoids mode collapse. A gradient penalty keeps the model well-behaved during training.

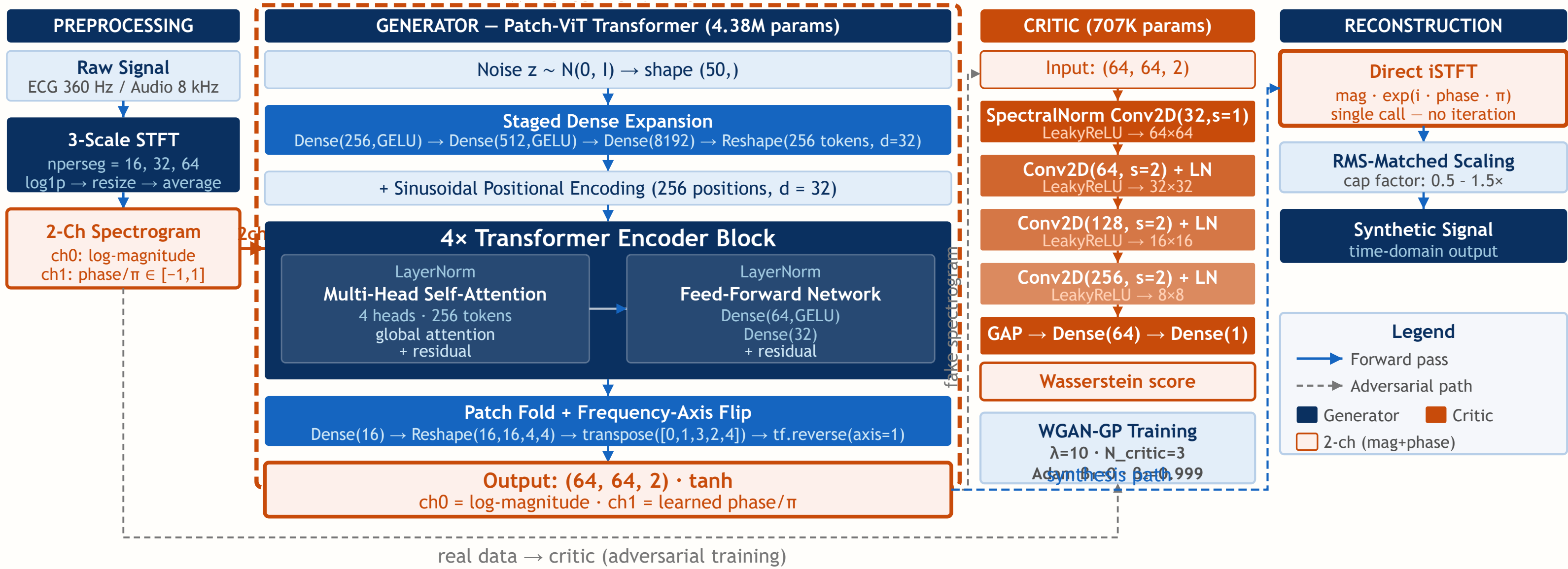
VISION TRANSFORMER / VIT (DOSOVITSKIY ET AL., 2021)

Splits the spectrogram into patches. Every patch attends to every other patch at once — capturing harmonic relationships that LSTM must propagate one step at a time.

DIRECT ISTFT RECONSTRUCTION **KEY NOVELTY**

Griffin-Lim guesses phase iteratively, smearing sharp features. Our generator outputs both magnitude and phase — reconstruction is a **single inverse STFT call**, preserving signal morphology exactly.

3 – PROPOSED ARCHITECTURE · SPECTRANS-ISTFT WGAN-GP



Aspect	Griffin-Lim (Combos A and C)	iSTFT Direct — our method (Combos B and D)
Generator output	Magnitude only (1 channel)	Magnitude + Phase (2 channels) ✓
Phase obtained by	32 iterative estimates	Learned directly from real data ✓
Reconstruction	Many inverse STFT calls	Single inverse STFT call ✓

4 – DATASET 1: MIT-BIH ECG

48 recordings · 3 classes: Normal / LBBB / RBBB	
Sampling rate	360 Hz native
Window per beat	256 samples = 711 ms
Train / Test	7,200 / 1,800
GAN seqs / class	2,000
Synthetic / class	100 (epochs: 2,000)

4 – DATASET 2: ESC-50 AUDIO

3 classes: Rain / Engine / Siren · 40 files × 5 s each	
Sampling rate	44,100 → 8,000 Hz
Window	128 samples = 16 ms
Train / Test	7,200 / 1,800
GAN seqs / class	2,000
Synthetic / class	100 (epochs: 2,000)

4 – 4 ARCHITECTURES TESTED

ID	Generator	Reconstruction
A	Transformer	Griffin-Lim
B ✓	Transformer	iSTFT direct
C	CNN	Griffin-Lim
D	CNN	iSTFT direct

Same critic and training for all four. **Combo B wins on both datasets.**

5 – RESULTS AND DISCUSSION

MIT-BIH ECG — ACCURACY

Combo	Gen	Recon	Accuracy	Δ
Baseline	Real only		95.11%	—
B ✓	Trans	iSTFT	95.89%	+0.78%
A	Trans	GL	95.50%	+0.39%
D	CNN	iSTFT	95.39%	+0.28%
C	CNN	GL	95.17%	+0.06%

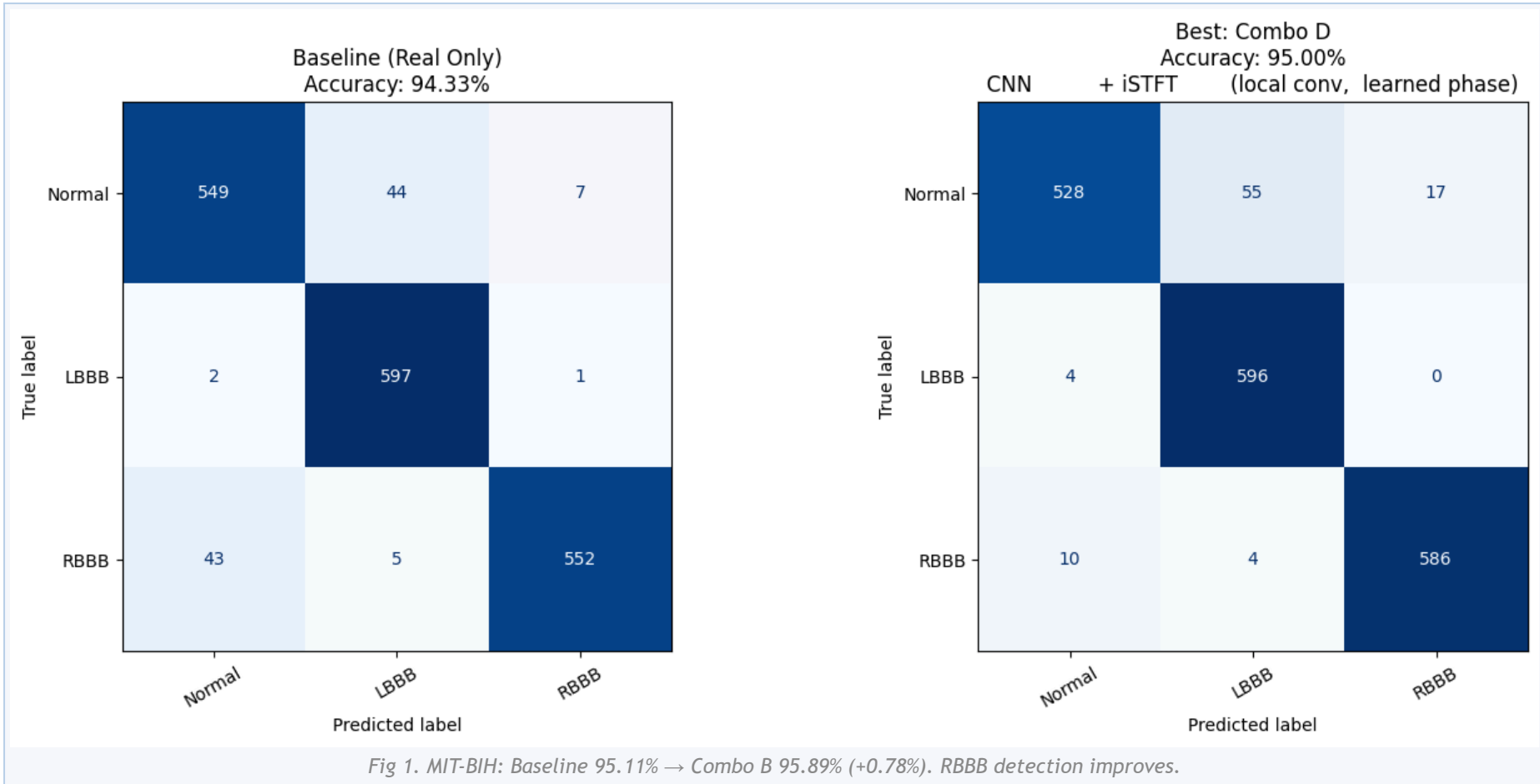
ESC-50 AUDIO — ACCURACY

Combo	Gen	Recon	Accuracy	Δ
Baseline	Real only		76.06%	—
B ✓	Trans	iSTFT	76.44%	+0.39%
A	Trans	GL	75.83%	−0.22%
C	CNN	GL	75.72%	−0.33%
D	CNN	iSTFT	75.50%	−0.56%

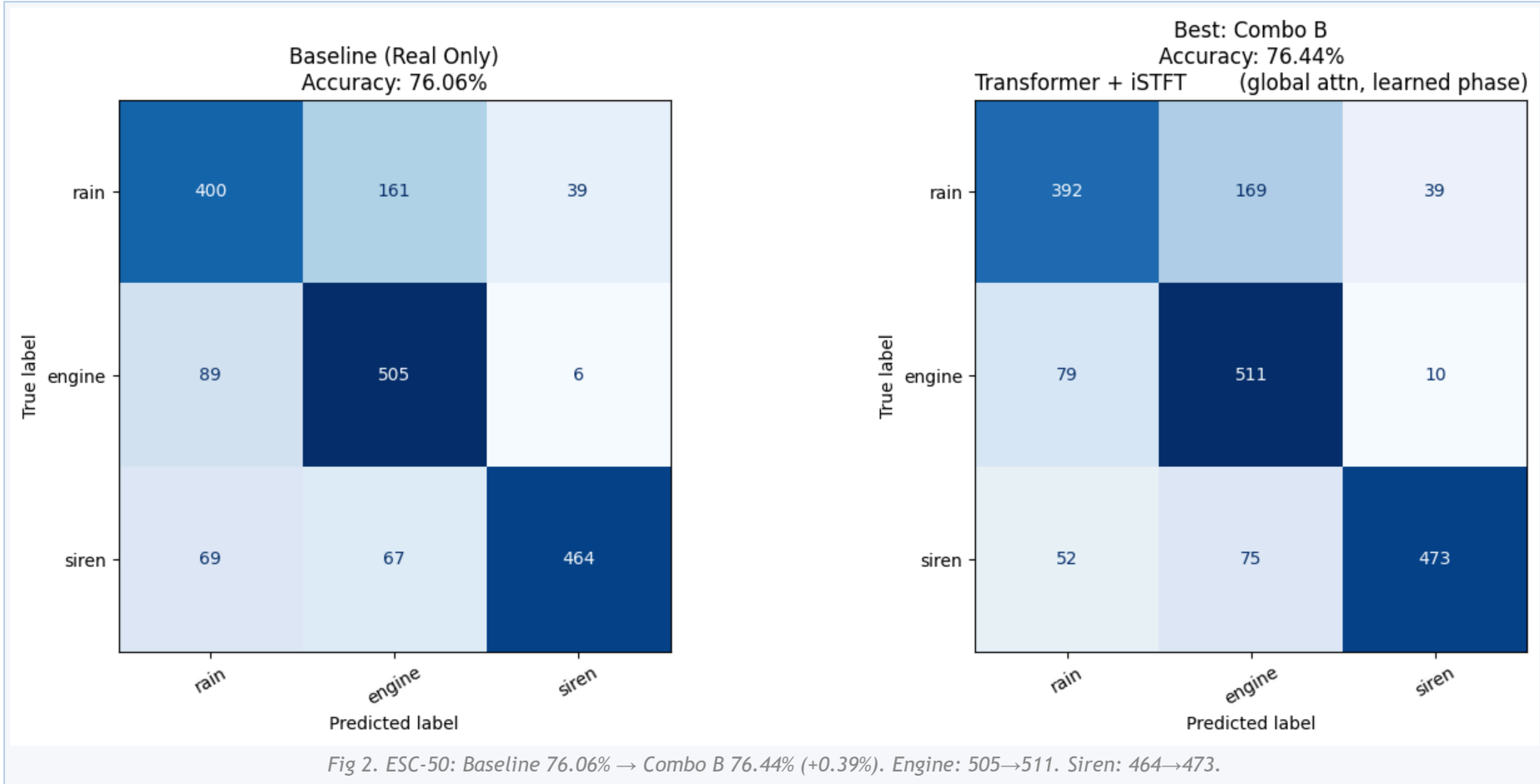
KEY FINDINGS

- Transformer + iSTFT consistently achieved the best performance.** Combo B was the only architecture that improved classification accuracy on both MIT-BIH ECG and ESC-50 audio, confirming that global self-attention with direct phase-aware reconstruction produces more realistic synthetic signals.
- Direct phase learning preserves waveform morphology.** Unlike Griffin-Lim, which estimates phase iteratively, iSTFT reconstruction directly predicts phase information, improving ECG morphology preservation and reducing spectral distortion.
- Transformer generators outperform CNN generators.** Vision Transformers captured long-range temporal-frequency dependencies more effectively than CNNs, especially for structured biomedical signals such as ECG beats.
- Synthetic ECG generation benefits more than environmental audio.** ECG improved more (+0.78%) because biomedical signals have consistent periodic structure, while environmental audio contains highly stochastic patterns.
- Multi-scale STFT improves representation quality.** Three STFT resolutions enabled the generator to learn fine-grained local patterns and broader frequency characteristics simultaneously.

MIT-BIH — BASELINE VS COMBO B (TRANS+ISTFT)



ESC-50 — BASELINE VS COMBO B (TRANS+ISTFT)



6 – CHALLENGES AND FUTURE WORKS

CHALLENGES

- Phase is circular:** Phase wraps at $\pm\pi$. Our tanh output does not handle this naturally, causing reconstruction errors at the boundary.
- Audio generation is harder:** Rain is random with no fixed pattern; siren pitch sweeps across time. The model produces structured spectrograms that do not fully capture this.
- Limited synthetic data:** 100 samples per class added to 2,400 real is a small fraction — limits measurable improvement on the full dataset.

FUTURE WORKS

- Circular phase loss:** Use a von Mises loss that understands phase wrapping. Expected to improve ECG beat sharpness and kurtosis match.
- More synthetic samples:** Generating 500–1,000 per class (vs 100) should produce a larger, more measurable accuracy improvement.
- Conditional generation:** Train the model to generate any class on request by including the class label as input — more practical for deployment.
- Longer training:** High Wasserstein distance at the end of training suggests the model had not fully converged. More epochs should improve signal quality.